

LEAD MAGNET | BOULDER BIOLABS

Chlorine Dioxide Sterilization Primer

Where CD fits, how the validation differs from EO, and what the Clordisys 2026 study means for your program.

Co-located with Boulder Sterilization. One ISO 13485 quality system. No shipping samples across the country.

Chlorine Dioxide Sterilization: A Manufacturer's Primer

Where chlorine dioxide fits, how the validation differs from EO, and what the Clordisys 2026 study means for your program.

Why Chlorine Dioxide?

Low temperature, low humidity.

Devices EO would damage (heat-sensitive electronics, bioabsorbable polymers, certain coatings) can tolerate CD.

Faster cycle than EO.

A typical CD cycle is measured in hours, not the 14-24 hours of an EO cycle.

No ECH or EG formation.

CD does not form the chlorinated or glycol byproducts EO does on chloride-bearing materials.

Spore kill at low concentrations.

Effective against *Geobacillus stearothermophilus* and other BIs at parts-per-million sterilant levels.

Trade-offs.

Smaller load chambers, residuals analytical chemistry less developed, fewer contract sterilizers offer it.

Validation Differences vs. EO

Step	EO Cycle	CD Cycle
Sterilant	Ethylene oxide gas	Chlorine dioxide gas
BI	B. atrophaeus	G. stearothersophilus or B. subtilis
Residuals standard	ISO 10993-7	Clordisys framework (March 2026)
Residuals analytes	EO, ECH, EG	ClO2, chlorite, chlorate
Residuals method	GC-FID	Ion chromatography + spectrophotometric
Toxicity thresholds	Patient-contact category	<6.83 mg/device chlorite, <70.65 mg/device chlorate
Contract sterilizers	Many	Few (only Boulder Sterilization in US under ISO 13485)
Contract residuals labs	Many	Few (Boulder BioLabs offers it)

How Boulder BioMed Operates the Full CD Stack

- **Boulder Sterilization:** CD cycles under ISO 13485 (the only U.S. contract sterilizer running them under that quality system).
- **Boulder BioLabs:** CD residuals testing aligned with the Clordisys framework. Ion chromatography for chlorite and chlorate.
- **Boulder RA/QA:** regulatory pathway support, from 510(k) framing through audit-readiness.
- **All co-located.** All under one ISO 13485 QMS, all on one Boulder, Colorado campus.

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